

Take 1:

Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Answer:

Procedure:

1. Wireshark was installed and opened on the laptop.
2. The active Wi-Fi network interface was selected.
3. Packet capture was started.
4. The website Instagram.com was opened in a web browser.
5. After the website loaded, the packet capture was stopped.
6. The Protocol column in the packet list was examined to identify the protocols.
7. The DNS packets were checked to identify the request for instagram.com.

Protocols Observed:

- DNS – Used to resolve instagram.com into an IP address.
- TCP – Used for reliable communication between the system and server.
- TLS – Provides encrypted and secure communication.
- QUIC – A modern transport protocol used by some web connections.
- UDP – Used as the transport protocol for applications such as QUIC.
- ARP – Used for resolving an IP address to a MAC address within the local network.

Task 2:

Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc).

Answer:

Packet 1 – DNS

OSI Layer: Application Layer (Layer 7)

DNS is used to translate a domain name such as instagram.com into its corresponding IP address.

Packet 2 – TCP

OSI Layer: Transport Layer (Layer 4)

TCP provides reliable and ordered delivery of data between the source and destination.

Packet 3 – TLS

OSI Layer: Presentation Layer (Layer 6)

TLS provides encryption and secure communication between the client and server.

Result:

Three packets were selected from the Wireshark capture and mapped to their corresponding OSI layers using the Protocol and Info columns.

Task 3:

Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

Answer:



1. The OSI Model consists of seven layers. Each layer performs a specific function during network communication.
2. Application Layer (Layer 7):
Example: WhatsApp
The Application Layer provides network services directly to users. Applications like WhatsApp use this layer for activities such as sending messages, sharing images, sending files, and making voice or video calls.

3. Presentation Layer (Layer 6):

Example: Instagram Images

The Presentation Layer is responsible for data formatting, encryption, and compression. For example, images and other data shared through Instagram need to be formatted and processed so that they can be properly displayed.

4. Session Layer (Layer 5):

Example: Google Meet

The Session Layer establishes, manages, and terminates communication sessions. In a Google Meet call, it helps maintain the communication session between the participants.

5. Transport Layer (Layer 4):

Example: TCP

The Transport Layer provides end-to-end communication between devices. TCP ensures that data is delivered reliably, in the correct order, and without missing important segments.

6. Network Layer (Layer 3):

Example: IP Address / Router

The Network Layer is responsible for logical addressing and routing. IP addresses are used to identify devices, and routers use this information to forward packets toward their destination.

7. Data Link Layer (Layer 2):

Example: Wi-Fi

The Data Link Layer handles communication between devices on the same local network. It uses MAC addresses and helps organize data into frames for transmission.

8. Physical Layer (Layer 1):

Example: Ethernet Cable / Wi-Fi Signals

The Physical Layer is responsible for transmitting raw bits over a physical medium. Data is transmitted as electrical signals through cables or as radio signals through wireless connections.

Task 4:

Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.

Answer:

How a WhatsApp Message Travels Through the OSI Model

When we send a message on WhatsApp, the message passes through different layers of the OSI Model before reaching the receiver. Each layer performs a different task to prepare and transmit the message.

1. Application Layer – Layer 7

This is the layer closest to the user. When we open WhatsApp and type a message such as “Hello”, the WhatsApp application processes our input and prepares it for sending. The Application Layer provides the services needed by the user to communicate through the application.

2. Presentation Layer – Layer 6

At this stage, the data is prepared in a suitable format for transmission. The data may be encoded, compressed, and encrypted. WhatsApp uses encryption to protect the message while it is being transmitted, so unauthorized people cannot easily read its contents.

3. Session Layer – Layer 5

The Session Layer manages the communication session between the sender and receiver. It helps establish and maintain the communication process while data is being exchanged. In simple words, it helps keep the communication session active while the message is being sent.

4. Transport Layer – Layer 4

The Transport Layer is responsible for end-to-end delivery of data. The message is divided into smaller pieces if necessary and transport protocols help deliver the data to the destination. This layer also helps manage the flow of data and communication between the sender and receiver.

5. Network Layer – Layer 3

The Network Layer is responsible for logical addressing and routing. The data is placed into packets and IP addresses are used to identify the source and destination. Routers on the Internet examine the addressing information and forward the packets toward the receiver's network.

6. Data Link Layer – Layer 2

The Data Link Layer handles communication over the local network. It adds information such as MAC addresses and converts the network-layer packet into a frame. The frame is then passed to the local network device, such as a Wi-Fi router.

7. Physical Layer – Layer 1

The Physical Layer is responsible for actually transmitting the data as signals. Depending on the connection, the data can travel as radio signals through Wi-Fi, electrical signals through a cable, or signals through other physical network media.

At the Receiver's Side

When the message reaches the receiver's device, the process happens in the reverse direction. The data moves from the Physical Layer up through the Data Link, Network, Transport, Session, Presentation, and Application Layers. Finally, WhatsApp processes the received data and displays the message on the receiver's screen.

Simple Flow

Type message → Prepare & encrypt → Manage session → Transport data → Route packets → Create frames → Send as signals → Receiver gets the message

Thus, the OSI Model helps explain how a WhatsApp message is prepared, packaged, transmitted through the network, and finally received and displayed on another device.